

IMO Training 2018–19 (Phase 1)
Level 1
Complex Numbers and Geometry

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July 19, 2018

1 Background knowledge

Problem 1 Discuss the Cartesian, polar and exponential forms of complex numbers. Also discuss the modulus, argument and conjugate of complex numbers.

Problem 2 Show that for any complex number z , we have $z\bar{z} = |z|^2$. Also, show that for any complex numbers z and w , we have $\bar{z} + \bar{w} = \overline{z + w}$ and $\bar{z} \cdot \bar{w} = \overline{z \cdot w}$.

Problem 3 Discuss the geometric meanings of addition, subtraction, multiplication and division of complex numbers.

Problem 4 Express the following geometric conditions in terms of complex numbers: (i) three points being collinear, (ii) two line segments being parallel, (iii) two line segments being perpendicular, (iv) four points being concyclic.

Problem 5 Suppose A , B and C are on the unit circle with center 0. Find the circumcenter, centroid, orthocenter and incenter of $\triangle ABC$.

2 Examples

Problem 6 (IMO 2012 problem 1) Given triangle ABC the point J is the centre of the excircle opposite the vertex A . This excircle is tangent to the side BC at M , and to the lines AB and AC at K and L , respectively. The lines LM and BJ meet at F , and the lines KM and CJ meet at G . Let S be the point of intersection of the lines AF and BC , and let T be the point of intersection of the lines AG and BC . Prove that M is the midpoint of ST .

(The excircle of ABC opposite the vertex A is the circle that is tangent to the line segment BC , to the ray AB beyond B , and to the ray AC beyond C .)

Solution. First we shall establish a lemma: If z and w are points on the unit circle with center 0, then the tangent passing through z and the tangent passing through w intersect at $\frac{2zw}{z+w}$. Indeed let the point of intersection be u . We have that $\frac{u-z}{z-0}$ is purely imaginary. Upon simplification this yields $u + z^2\bar{u} = 2z$. Similarly $u + w^2\bar{u} = 2w$. Solving, we get $u = \frac{2zw}{z+w}$. (In class I misstated the result as $\frac{zw}{z+w}$, i.e. I missed the factor 2. As a result the subsequent calculations were erroneous. Please accept my apologies.)

Now we turn to the main problem. Let the said excircle be the unit circle with center 0, i.e. $j = 0$. Also let $m = 1$. Then by the above lemma, we have $b = \frac{2k}{k+1}$, $c = \frac{2l}{l+1}$ and $a = \frac{2kl}{k+l}$.

Now we calculate f . We have that $\frac{f-0}{b-0}$ is a real number, and $\frac{f-1}{l-1}$ is a real number. Solving, $f = \frac{kl+k}{k+l}$. Similarly $g = \frac{kl+l}{k+l}$.

Now we calculate s . We have that $\frac{s-a}{f-a}$ is a real number and $\frac{s-1}{1-0}$ is purely imaginary. Solving, $s = \frac{2k}{k+l}$. Similarly $t = \frac{2l}{k+l}$. So $\frac{s+t}{2} = 1$ as desired.

Problem 7 (IMO 2018 problem 1) Let Γ be the circumcircle of acute triangle ABC . Points D and E are on segments AB and AC respectively such that $AD = AE$. The perpendicular bisectors of BD and CE intersect minor arcs AB and AC of Γ at points F and G respectively. Prove that lines DE and FG are either parallel or they are the same line.

Problem 8 (CMO 2012 problem 1) In $\triangle ABC$, $\angle A$ is the largest. On the circumcircle of $\triangle ABC$, let D be the midpoint of arc $\widehat{ABC}$ and E be the midpoint of arc $\widehat{ACB}$. The circle c_1 passes through A, B and is tangent to AC at A , the circle c_2 passes through A, E and is tangent AD at A . The circles c_1 and c_2 intersect at A and P . Prove that AP bisects $\angle BAC$.